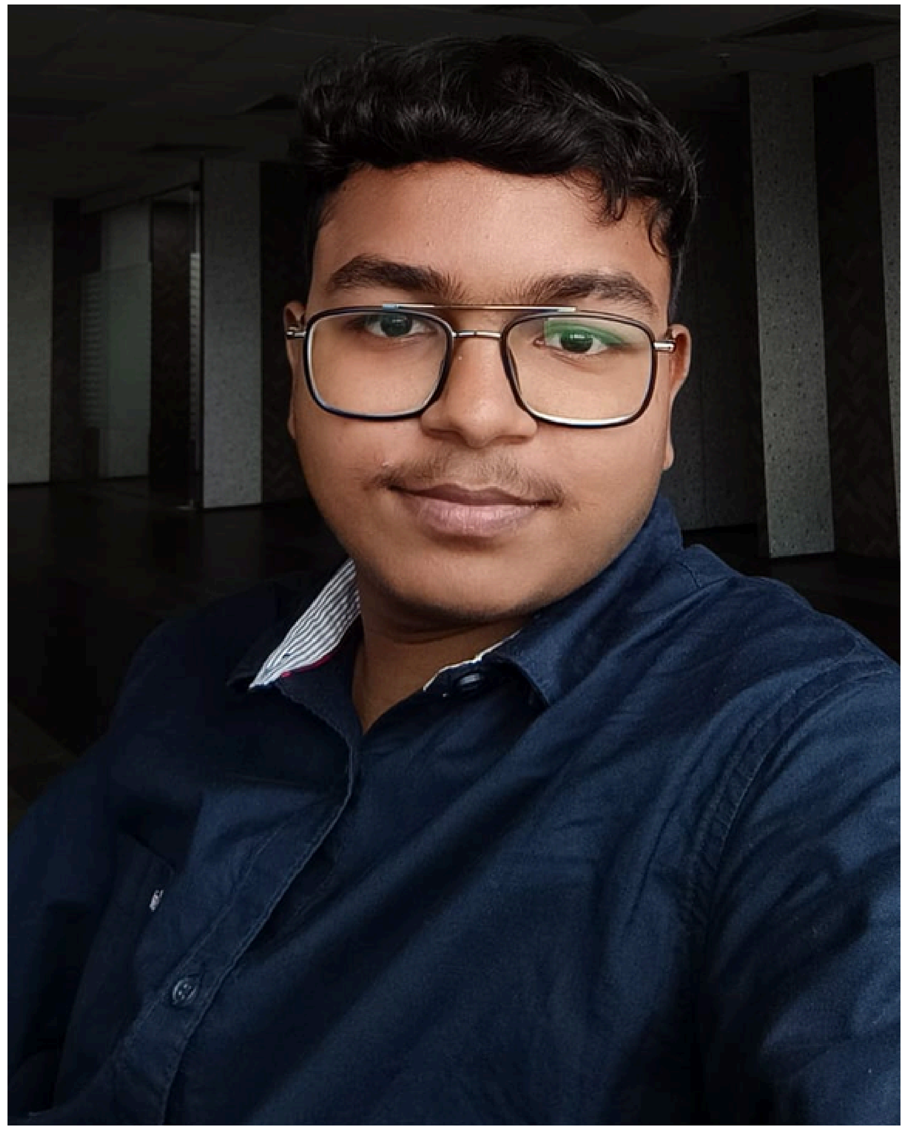
AI-based Reading & Digitization of Handwritten Prescriptions

HEALTH DEPARTMENT, COE DS&ML

5TH MARCH 2026 - 5TH JUNE 2026

Our Team



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Problem Statement

1. Handwritten prescriptions are difficult to interpret (poor handwriting, inconsistent formats, abbreviations)

2. Leads to:
   - Time consumption in healthcare processes
   - Lack of proper digital records
   - Difficulty in analysis of data

Goal

Convert handwritten prescriptions into accurate digital records using AI

- To develop an AI system to read handwritten prescriptions
- To extract and structure key medical information
- To reduce errors and improve healthcare data management
- To store healthcare data

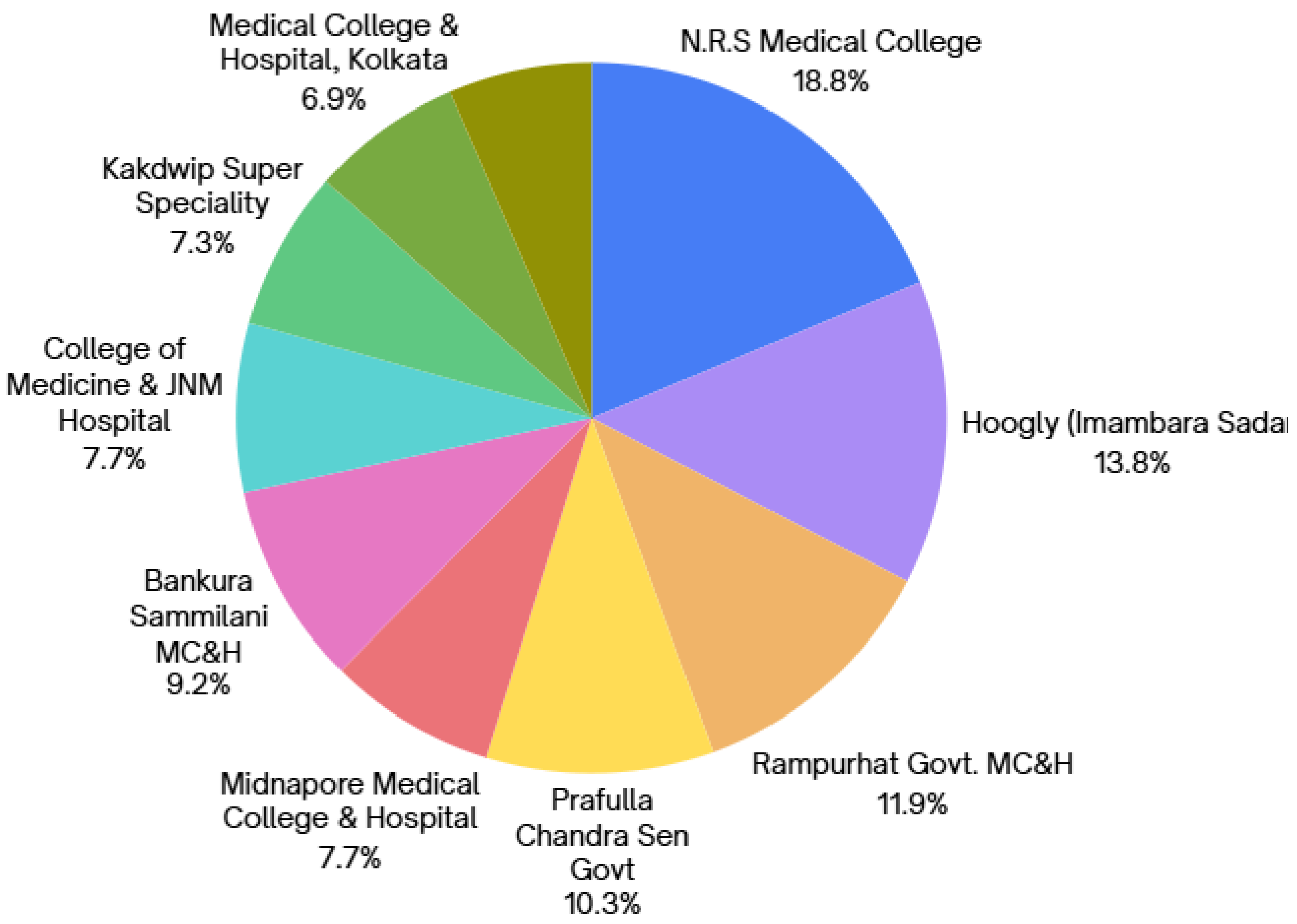
Project Contributions

Completed Deliverables

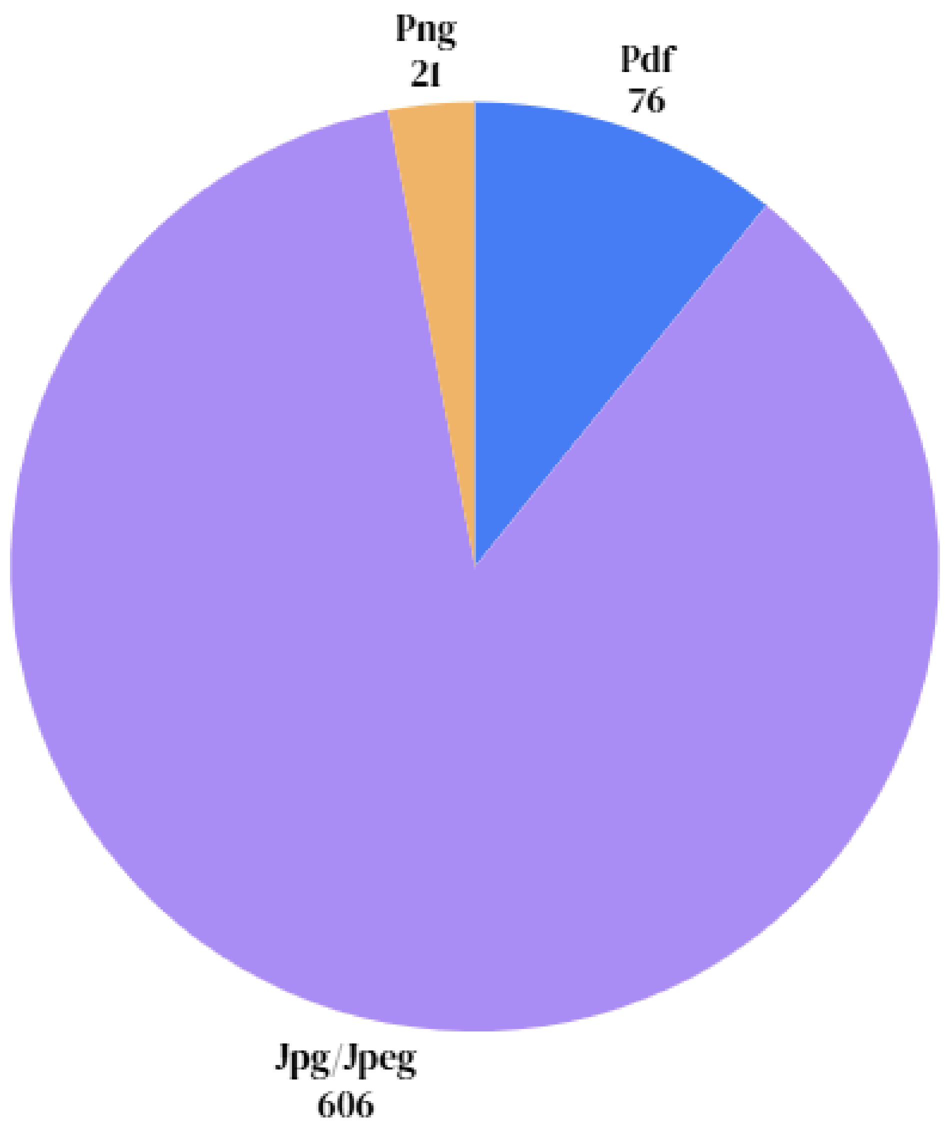
✓ Requirement Analysis
✓ Dataset Analysis
✓ YOLO-based PII Detection
✓ Automated Anonymization Pipeline
✓ Multimodal LLM Extraction

✓ Structured JSON Generation
✓ Domain Vocabulary Grounding
✓ Hybrid Actor–Critic Evaluation
✓ Evaluation Dashboard
✓ Research Paper Preparation

Data Analysis (EDA)

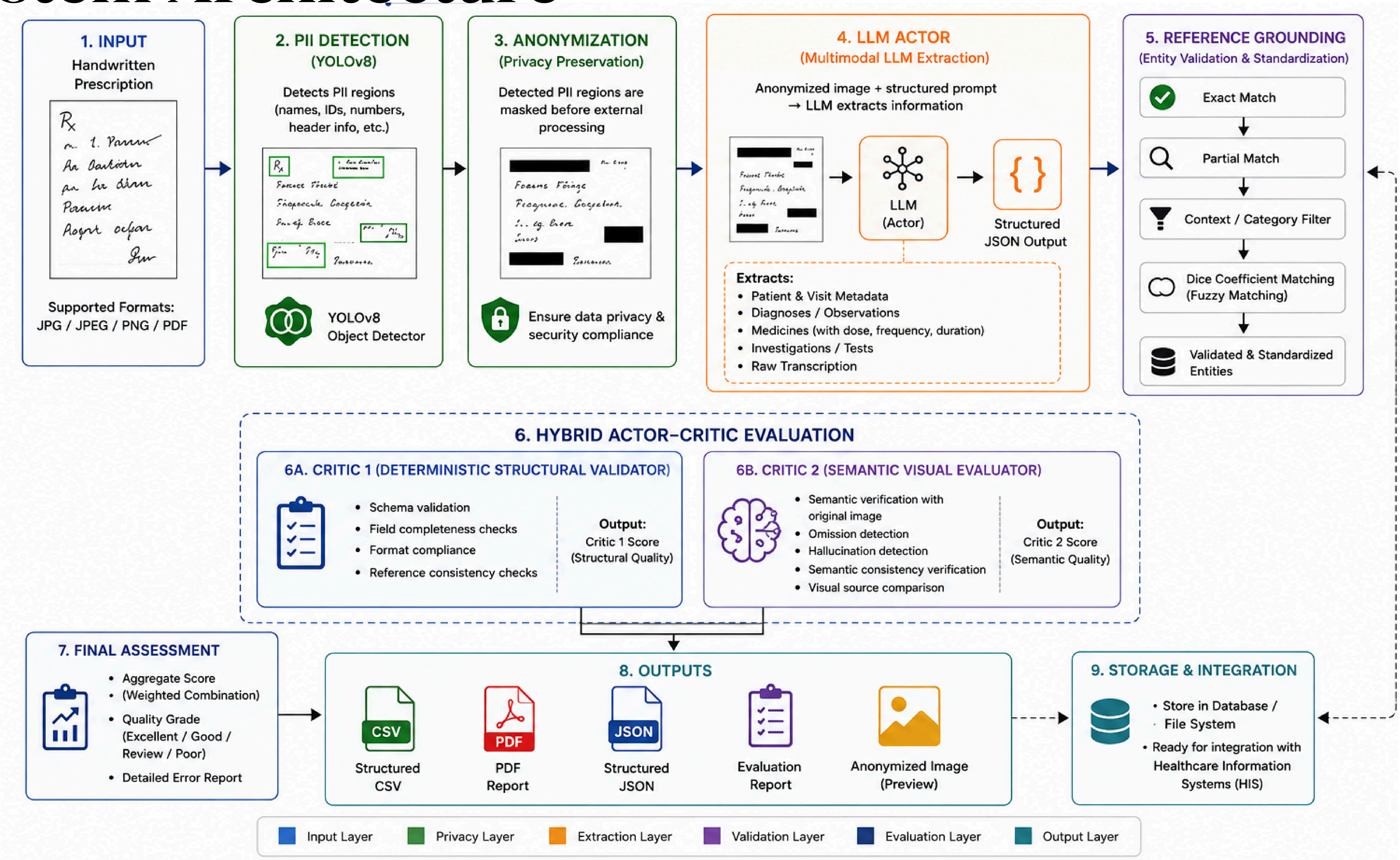


- Dataset includes prescriptions from 64 hospitals in total
- The pie chart represents top 10 hospitals based on maximum prescription count.



- Dataset contains prescriptions in multiple file formats:
  PDF: 76 prescriptions
  JPG/JPEG: 606 prescriptions
  PNG: 21 prescriptions

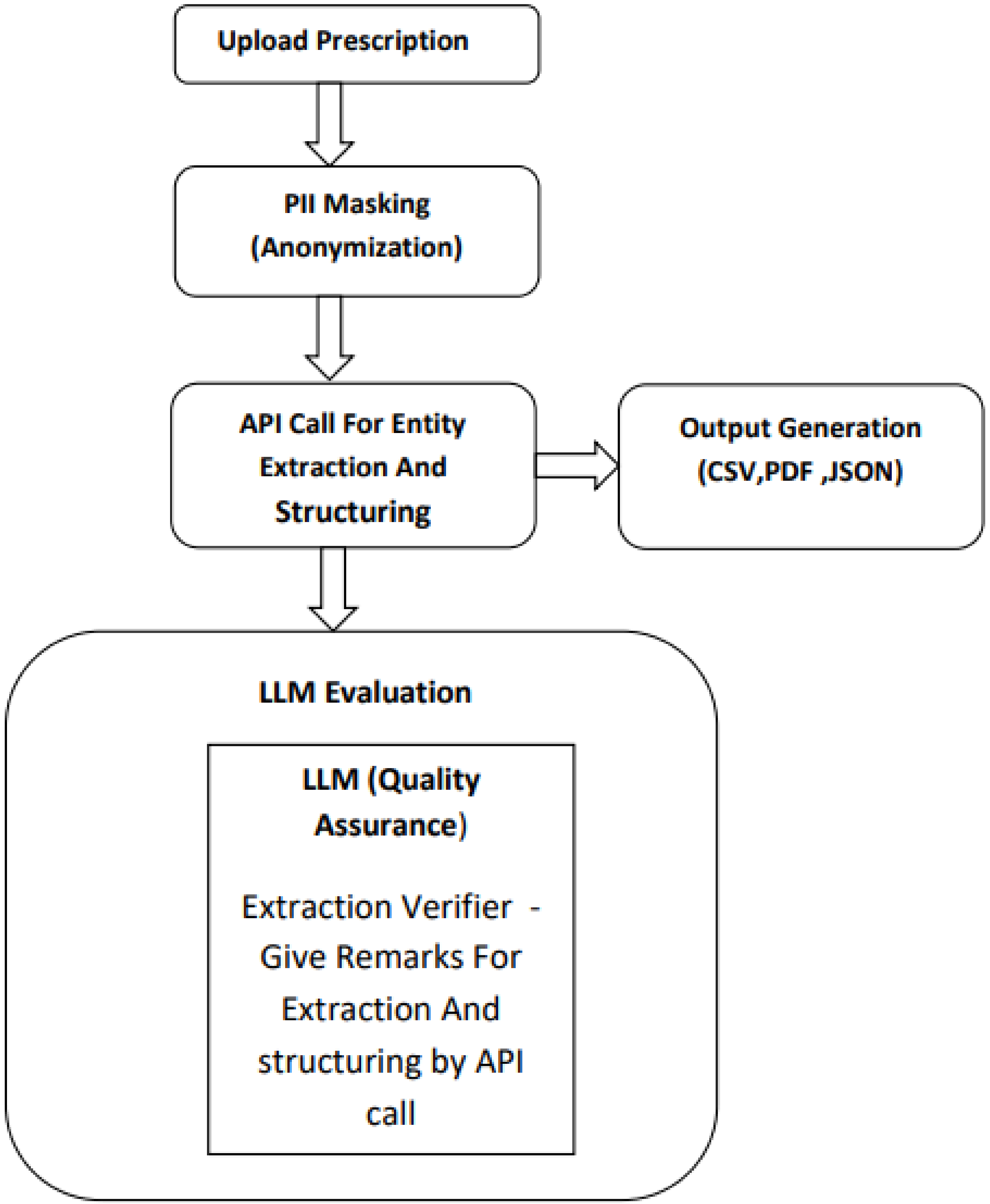
System Architecture



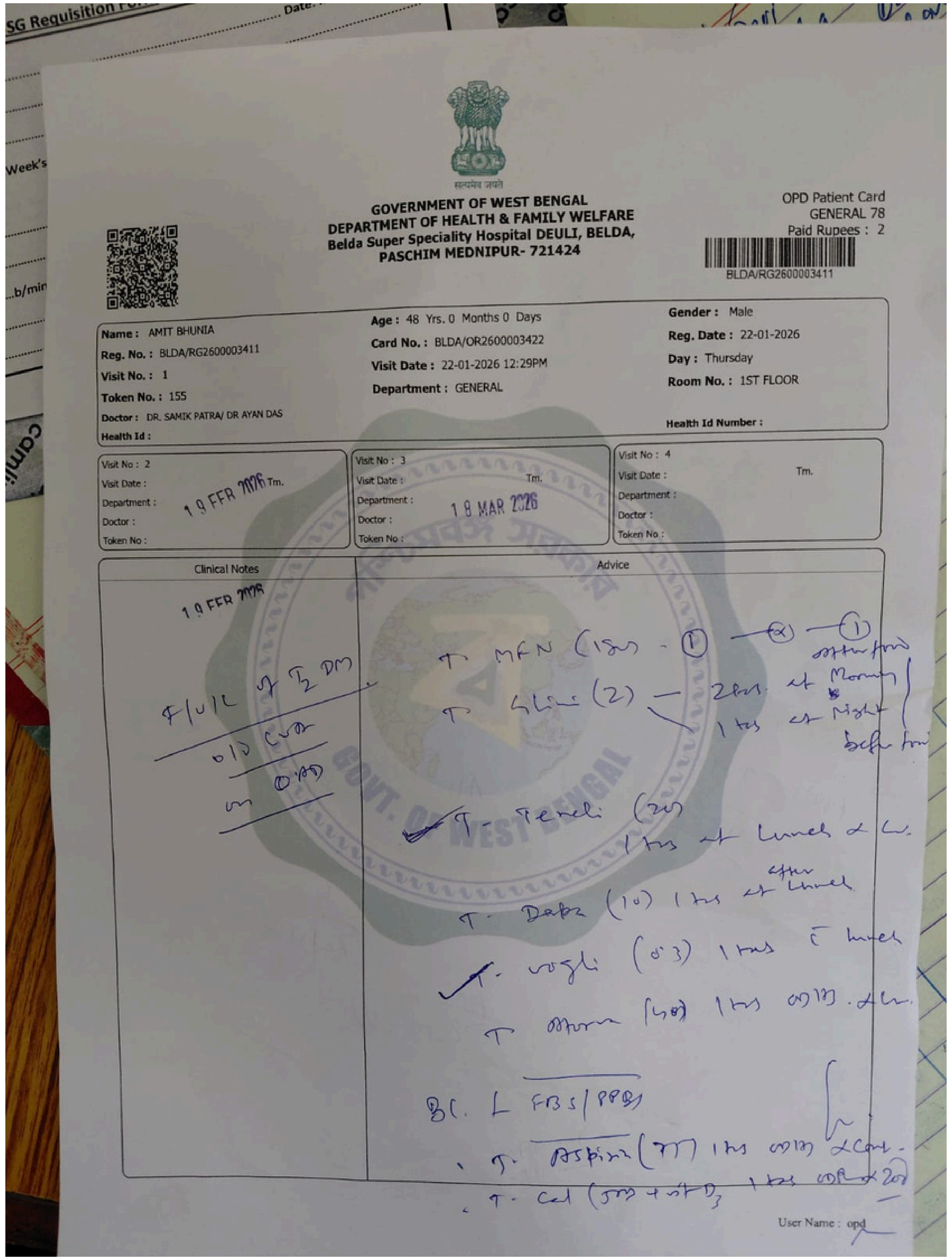
Workflow Demonstration

Process Overview

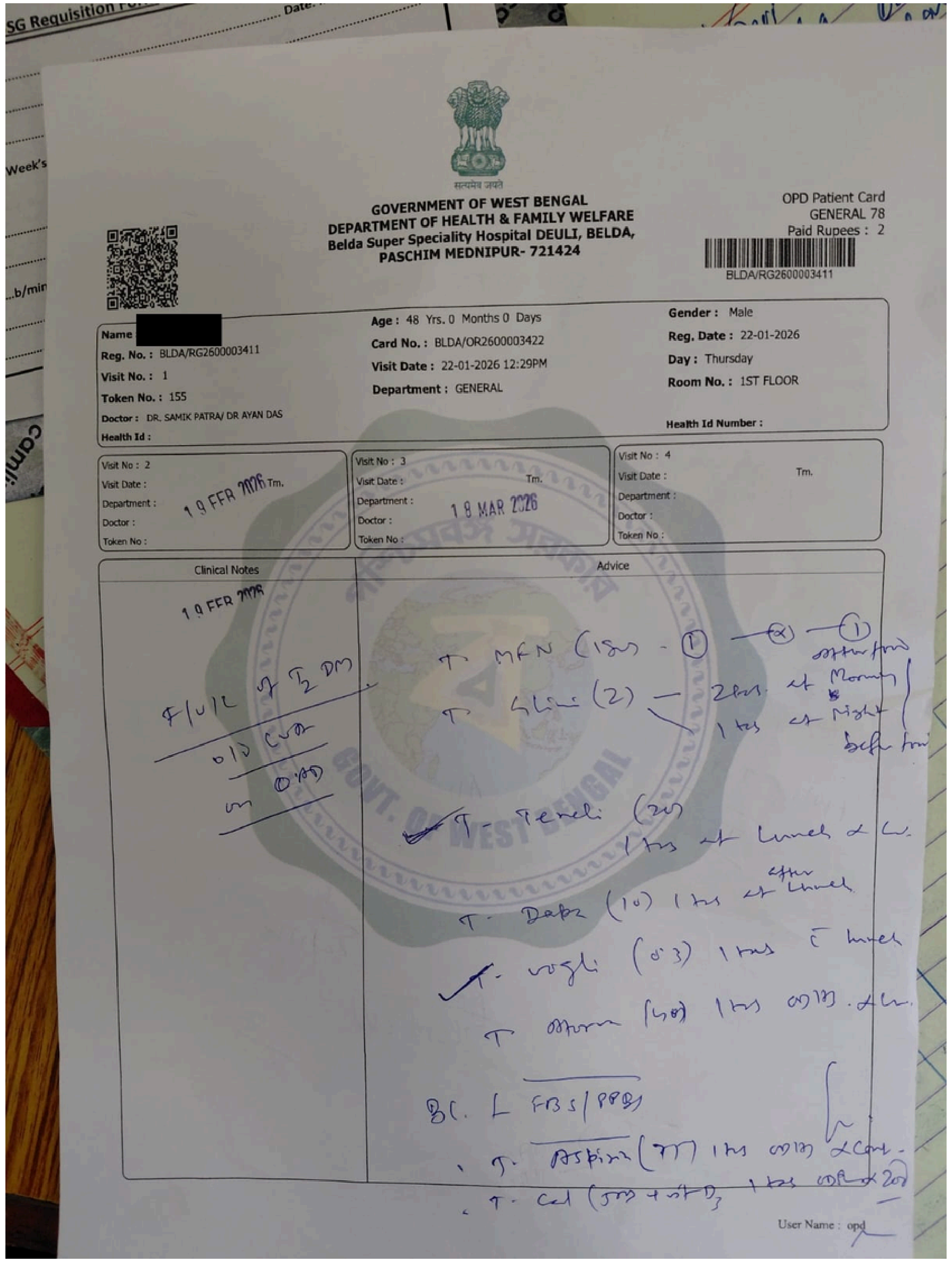
- End-to-end automated prescription processing
- Privacy ensured through anonymization
- Accurate extraction using API
- Output generation (CSV,PDF,JSON)
- LLM-based evaluation for validation



Privacy - Preserving Anonymization

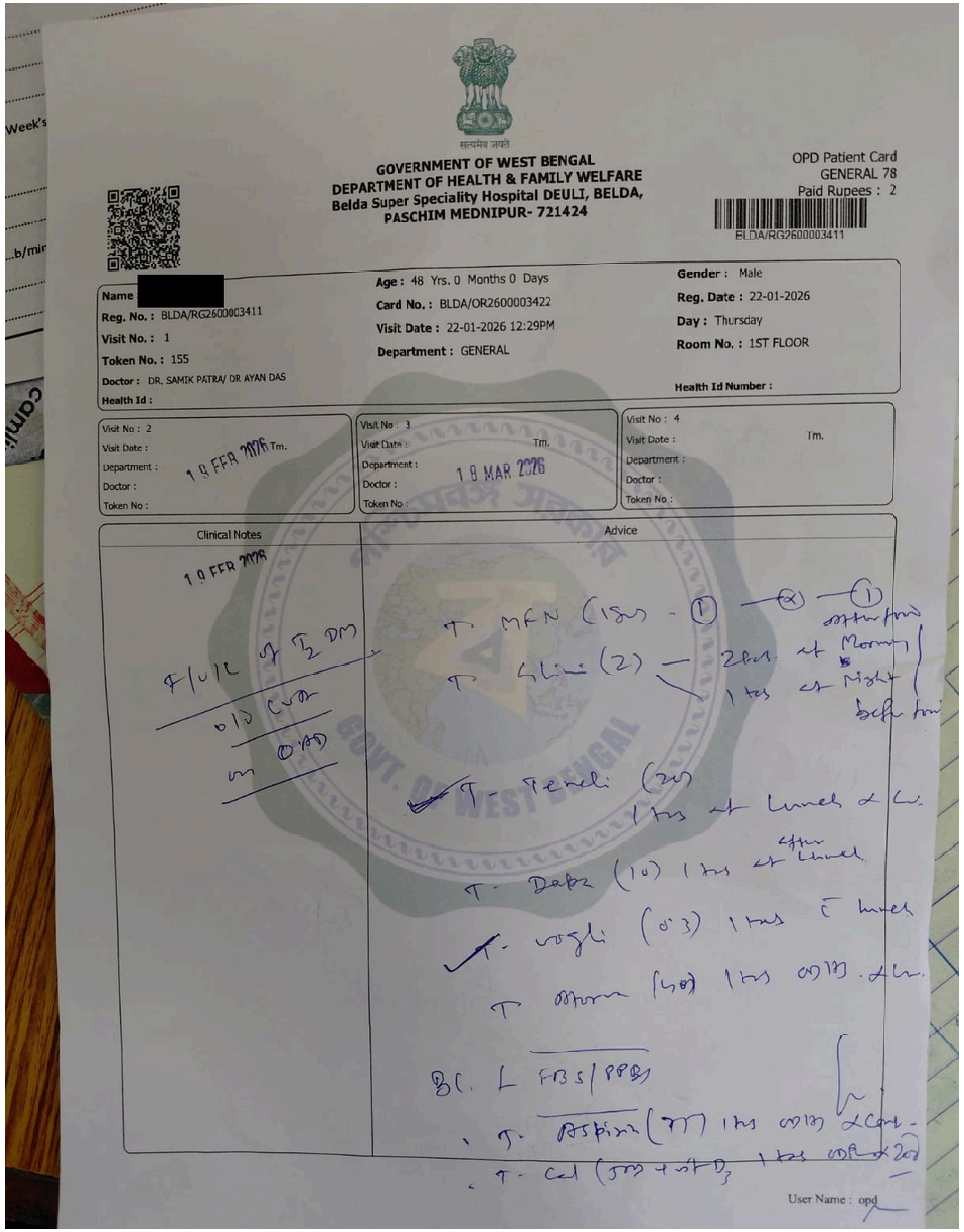


Original Prescription (Unmasked)



Masked Prescription (PII Removed)

Sample Input: Handwritten Prescription



GOVERNMENT OF WEST BENGAL
DEPARTMENT OF HEALTH & FAMILY WELFARE
Belda Super Speciality Hospital DEULI, BELDA,
PASCHIM MEDNIPUR- 721424

OPD Patient Card
GENERAL 78
Paid Rupees : 2
BLDA/RG2600003411

Name [redacted]	Age : 48 Yrs. 0 Months 0 Days	Gender : Male
Reg. No. : BLDA/RG2600003411	Card No. : BLDA/OR2600003422	Reg. Date : 22-01-2026
Visit No. : 1	Visit Date : 22-01-2026 12:29PM	Day : Thursday
Token No. : 155	Department : GENERAL	Room No. : 1ST FLOOR
Doctor : DR. SAMIK PATRA/ DR AYAN DAS		Health Id Number :
Health Id :		

Visit No : 2	Visit No : 3	Visit No : 4
Visit Date : 19 FEB 2026 Tm.	Visit Date : 18 MAR 2026 Tm.	Visit Date : Tm.
Department :	Department :	Department :
Doctor :	Doctor :	Doctor :
Token No :	Token No :	Token No :

Clinical Notes	Advice
19 FEB 2026 F/U/C of T2 DM o/c CVD	T. MFN (1500) T. Glim (2) T. Teneli (20) T. Dapa (10) BL FBS/PPBS T. Aspirin

User Name : opd

Extracted Output (Structured Data)

Digitized Patient Record

CSV | JSON | Save PDF

PATIENT DEMOGRAPHICS

Name: **Not verifiable**	Age/Sex: **32 Yrs. / Male**
Reg/Health ID: **DADH/RG2600024097**	Latest Visit Date: **18-Mar-2026**
Token / Room: **Not verifiable / 15**	Doctor: **Dr. Rinchen Lama**

VITALS & CLINICAL NOTES

Chief Complaints:

Not verifiable

Other Findings / Notes:

CONTI MEDICINE

ADVISED INVESTIGATIONS / SCANS

- **Not verifiable**

PRESCRIBED MEDICATIONS

Carboxy Methyl Cellulose Sodium

H04.1

✍ Written as:

`1. Carboxy Methyl Cellulose Sodium (CMC) 0.5% Eye Drops 30DAYS`

0.5%

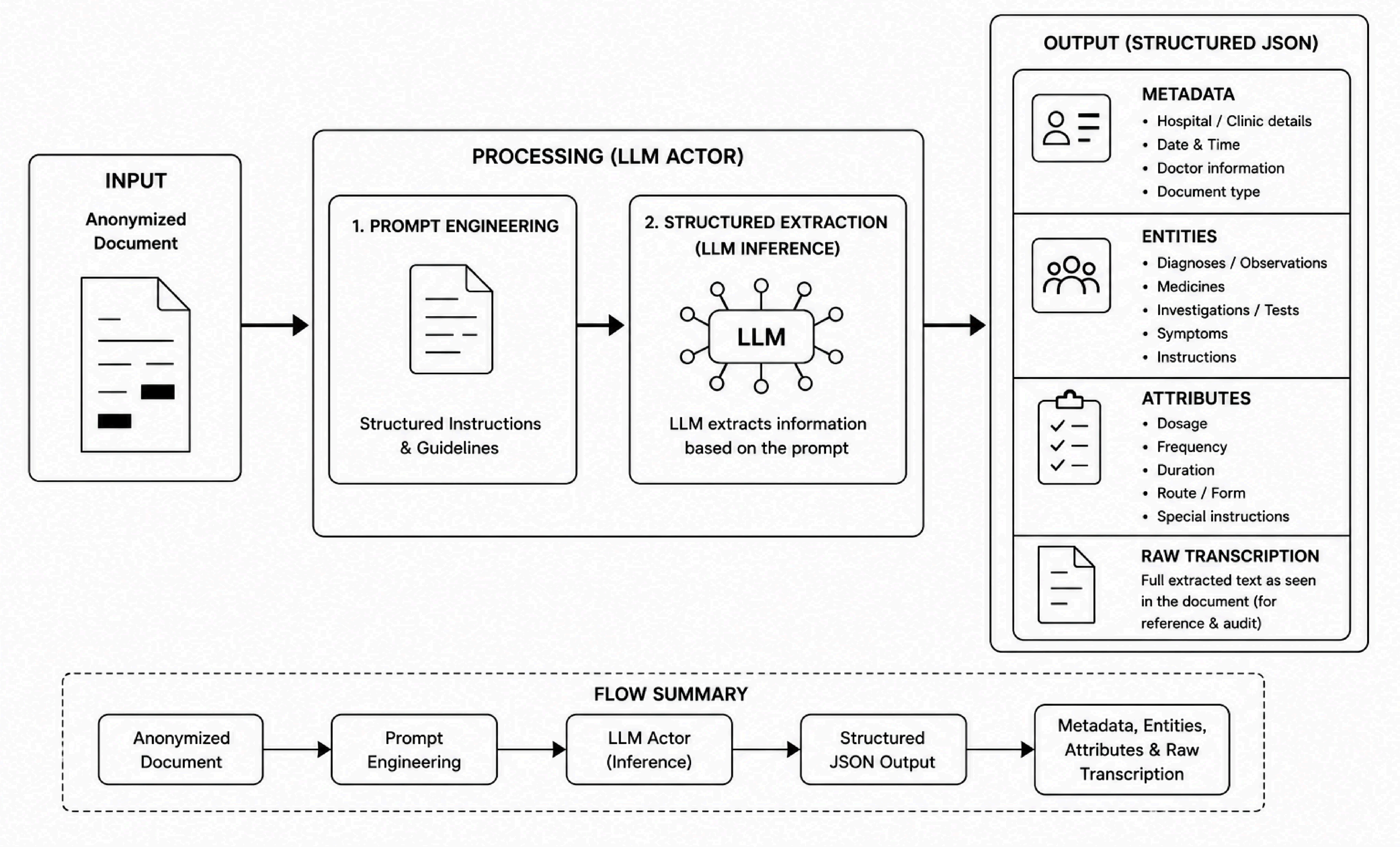
✅ **CMS Verified Match: Carboxy Methyl Cellulose Sodium (CMC) 0.5% Eye Drops**

Frequency: **30 days**

Instructions: Eye Drops

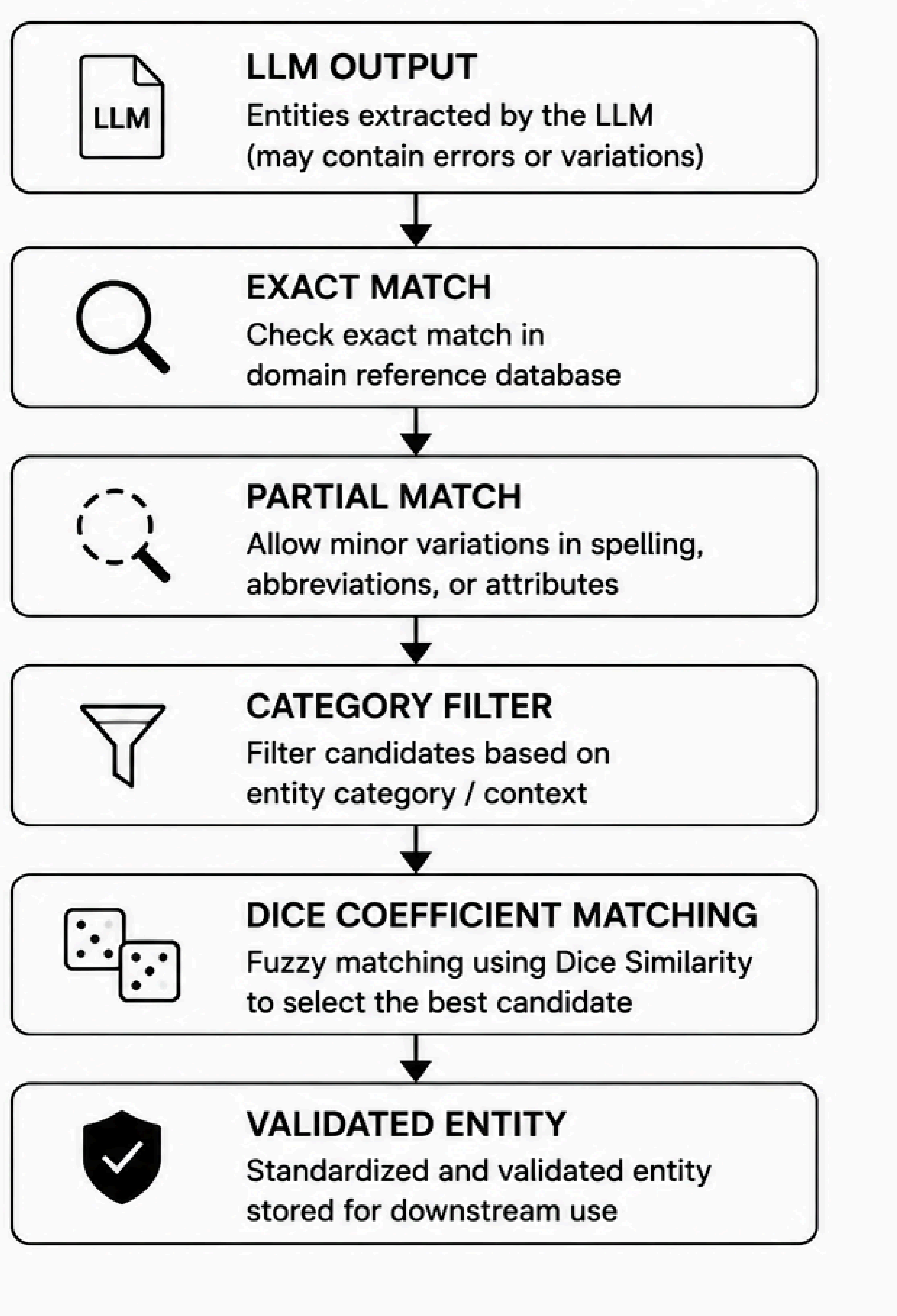
Actor (LLM Extraction)

- Processes anonymized handwritten documents.
- Performs prompt-guided structured extraction.
- Generates machine-readable JSON output.
- Preserves both structured data and raw transcription.

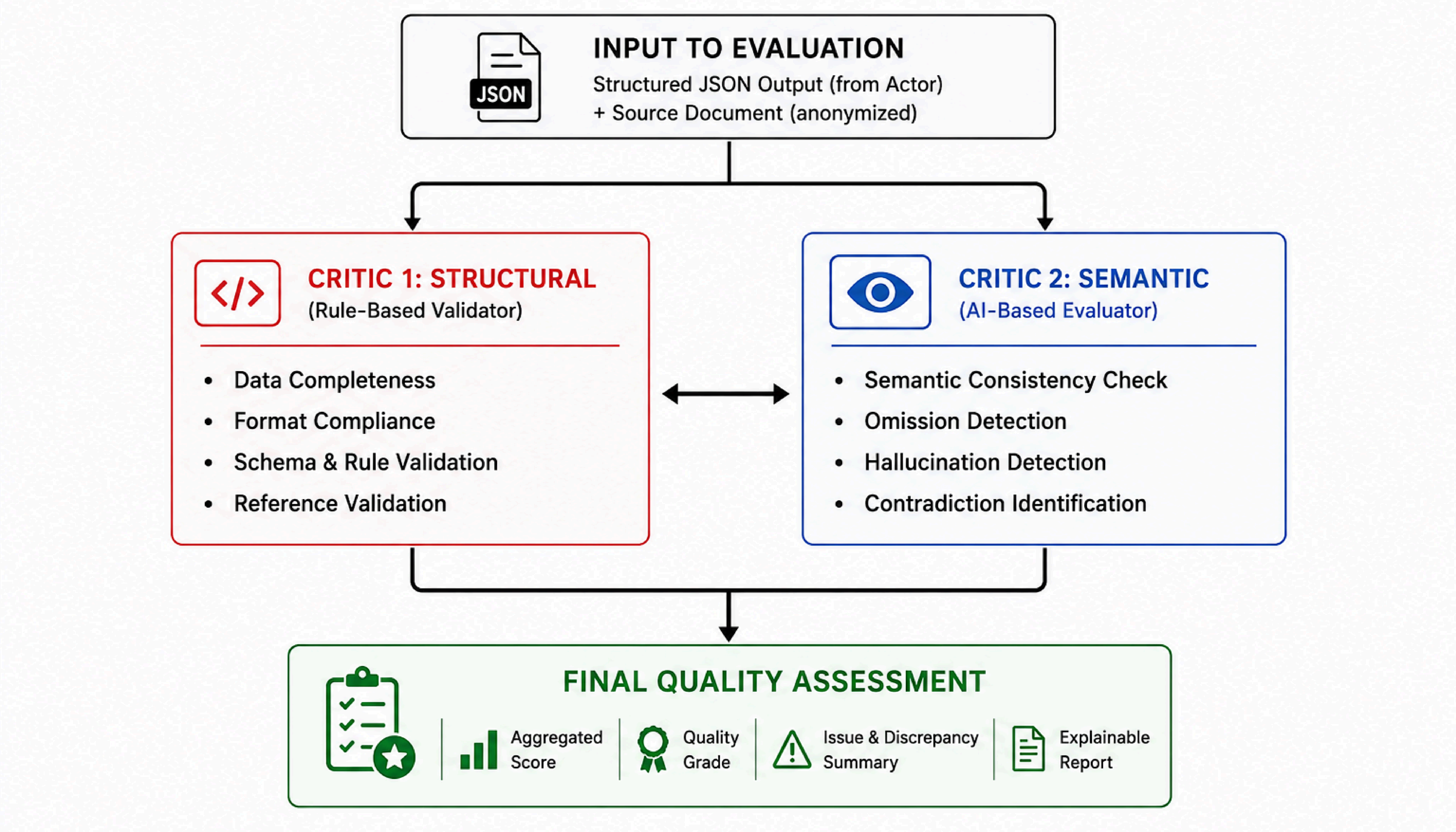


Domain Grounding

- Validates extracted entities using a reference database.
- Performs exact, partial, and similarity-based matching.
- Uses Dice Coefficient matching for noisy extractions.
- Standardizes terminology and improves consistency.
- Reduces hallucinations and improves extraction reliability.



Hybrid Actor-Critic Evaluation Framework



LLM Evaluation Output

Extraction Quality Score

Excellent

92.5/100

LAYER 1: STRICT REGEX AUDITOR

CMS Mapping:	**15/20**
ICD-10 Format:	**15/20**
Demographics:	**0/10**

LAYER 2: SEMANTIC AI CRITIC

All Meds Found:	**15/15**
All Notes Found:	**0/10**
Instruction Acc:	**0/10**
Hallucinations:	**7.5/15**

- [Fatal Hallucination - Entity]: H04.1

APPENDIX : System workflow

- The system takes a scanned handwritten prescription as input.
- Personal information is masked (PII anonymization) to ensure data privacy and security.
- The prescription image is processed using OCR via API to extract textual content.
- The extracted text is then passed to an LLM for entity extraction, including: Symptoms , diagnosis , medicines with dosage and frequency , recommended tests.
- The extracted information is converted into a structured format (JSON) and further exported as CSV/PDF.
- The output is evaluated using an LLM as a Judge to assess accuracy and completeness.

Note

- The system is designed to handle unstructured and diverse prescription formats.
- If the extracted output does not match expected results, an ***agentic system or Human-in-the-Loop (HITL) approach*** can be incorporated for validation and correction.
- This ensures higher reliability and readiness for real-world healthcare deployment.

Conclusion

- Developed an end-to-end framework for handwritten document digitization.
- Integrated privacy-preserving anonymization, LLM-based extraction, entity grounding, and hybrid Actor–Critic evaluation.
- Improved extraction reliability through exact, contextual, and Dice Coefficient–based matching.
- Demonstrated a scalable architecture that can be adapted to healthcare, legal, administrative, and archival document domains through configuration changes rather than code modifications.

This solution enhances accuracy, accessibility, and efficiency in healthcare data management

~ THANK YOU ~